



Technical Service Bulletin: TSB210096

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Exhaust Valve Insert Material Change for Industrial, Construction, and Powergen Applications

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

Engines:

- QSK45 CM500
- QSK60 CM2150 MCRS
- QSK60 CM500
- QSK60 CM850 MCRS
- QSK60 CM2250 K112

Applications:

- Tier 1 and Tier 2 engines in industrial/construction applications
- G8 engines in Powergen application

Description of Change

All industrial and construction applications are to moving from cylinder heads incorporating exhaust valve seats/inserts manufactured from J3 materials to J10 material. The two new part numbers also incorporate the improved casting for the cylinder heads detailed in Technical Service Bulletin, New Cylinder Head with Improved Casting, TSB210043.

Reason for Change

- Reduce valve seat drops
- J10 material has been selected to replace J3 material as the exhaust valve insert material due to increased thermal properties under high peak cylinder pressure and peak cylinder temperature conditions.

Service Instructions

- This document does **not** authorize proactive field repairs.

- In the event of a malfunction requiring the installation of a new cylinder head or cylinder heads, the new part number is to be used. See Table 1 below.
- The new part number should be applied in full to engines being rebuilt.

Service Parts Availability

Service parts are available. See Table 1 for part numbers.

Table 1, Service Parts

Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
Cylinder Head (Tier 1)	5371843	Yes	Yes	5635343
Cylinder Head (Tier 2)	5372056	Yes	Yes	5635345
Cylinder Head	5538438	No	No	5635343

Part Compatibility

The existing part number and new part number can be mixed on the same engine.

Part Inventory Action

Use the existing part number in applications that are **not** relevant to the application for which this TSB is being released.

Production Status

Implemented for production. See Table 2.

Table 2, Production Information

Part Description	New Part Number	ESN First	Build Date ¹	Plant
Cylinder Head (Tier 1)	5635343	33224355	25 March 2021	Daventry, UK
Cylinder Head (Tier 2)	5635345	33224317	22 March 2021	Daventry, UK
Cylinder Head	5635343	85010768	12 October 2021	CTIL, India

Document History

Date	Details
2021-5-5	Module Created
2021-5-12	Updated part numbers in Table 1.
2021-10-18	Added QSK60 CM2250 K112.
2021-11-29	Added QSK45 CM500.

